

Blockchain Constraints

Smart Contracts Do Not Run Automatically

Ethereum smart contracts execute only when a transaction is sent. This means:

- The contract cannot update itself automatically.
- Time-based decay cannot happen continuously.
- State changes occur only when a user interacts.

As a result, reputation decay is calculated when a user interacts after inactivity. This is a fundamental **limitation** of blockchain execution.

Timestamp Limitations

The system uses block.timestamp to measure time. However:

- Timestamps are set by block validators.
- They are approximate.
- They can vary slightly within acceptable limits.

Therefore:

- The system is designed for coarse intervals (e.g., 10 days).
- It does not rely on precise second-level accuracy.
- This makes the decay model stable and resistant to minor timestamp variations.

Gas Costs

Smart contract operations consume gas. This affects design decisions:

- Explanations are emitted as events instead of stored permanently.
- The logic is kept simple to reduce computational cost.
- Reputation updates avoid unnecessary loops or heavy calculations.

Immutability Risks

Blockchain data is immutable. Meaning→

- Logic errors cannot be corrected after deployment without redeploying a new contract.
- Mistakes in rule design become permanent.

To address this→

- The contract will be deployed on a test network.
- The logic is carefully reviewed before deployment.

No Off-Chain Enforcement

The system does not verify →

- Real-world identity,

- Authenticity of behavior,
- Whether interactions are meaningful.
- (for now, impact based logic off the scope)

Scalability Limitations

The system is not optimized(for now) for:

- Large-scale production use,
- Millions of users,
- High-frequency interaction environments.

“It is designed as a prototype.”

External Dependency

The frontend depends on →

- A web browser,
- MetaMask or some wallet,
- Ethereum test network availability.

If these components are unavailable, the interface cannot function, although the contract remains deployed.